

A Joint Bandwidth Expansion and Speech Enhancement Approach using Deep Neural Network

Taieba Taher, Nursadul Mamun, Md. Azad Hossain

Dept. of Electronics and Telecommunication Engineering, Chittagong University of Engineering and Technology, Bangladesh
(taieba.athay; nursad.mamun; azad)@cuet.ac.bd

Abstract— Recently, joint bandwidth expansion and speech enhancement has been a topic of interest in the field of speech processing. The main challenge in this task is to increase the bandwidth of speech signals while enhancing their quality, simultaneously. Deep neural networks (DNNs) have shown great promise in addressing this challenge, as they can learn complex relationships between the input and output signals. In this study, a joint bandwidth expansion and speech enhancement approach using DNNs have been proposed, which is designed to simultaneously increase the bandwidth of speech signals and reduce noise, while preserving speech quality and intelligibility. This approach leverages the capability of DNNs to simultaneously estimate the missing speech components and the noise profile in the degraded speech signal. The estimated speech components and the noise profile are then used to synthesize a full-band speech signal from a noisy signal with limited bandwidth with improved quality. The network employs three different phases such as oracle, imaged, and noisy phase along with the magnitude spectra to recover high band components. The joint approach demonstrates that the DNN-based bandwidth extension and speech enhancement can be effectively combined to produce high-quality speech signals, outperforms traditional speech enhancement methods, and offers promising solutions for various applications, including speech communication, speech recognition, and speech synthesis.

Keywords—Speech communication, Speech intelligibility, DNN, Speech Enhancement, Bandwidth Extension, Speech Quality.

I. INTRODUCTION

Bandwidth extension is a technique utilized in audio signal processing to enhance the quality of audio signals. This method broadens the frequency range of the original audio signal, resulting in a more vivid and complete sound. In numerous practical applications, the quality of speech signals is impaired by various forms of noise, making speech enhancement a vital task. Bandwidth extension is one solution to enhance the quality of speech signals, as it expands the bandwidth while still maintaining their intelligibility. A joint approach is proposed to mitigate the impact of bandwidth limitations in communication systems, such as telephone networks, and enhance the quality of speech signals for applications like speech recognition and speaker identification. Speech improvement with bandwidth extension may be classified into two approaches: Approaches based on models and approaches based on data. Approaches based on model utilize mathematical models of speech signals to generate new spectral information, while approaches based on data leverage statistical models trained on speech data to estimate and generate new spectral information. Both approaches have their own advantages and limitations, and the appropriate approach is determined based on the specific needs of the application. Several methods have been employed till now to facilitate the retrieval of speech for

recipients [1, 2]. Two primary branches of bandwidth extension have been proposed in [1]. One of these approaches utilizes the only available low-frequency information to extract high frequencies whereas other methods use auxiliary information along with the available low-frequency spectra [2].

Source filtering approaches are utilized to recover the high-frequency range from the residue of low-frequency signal using simple modulation techniques like spectral folding [3] with the estimation of the high-frequency spectral envelope which is rather a challenging task and the excitation created from the low-frequency component. For constructing the connection between high-frequency spectral envelope and low-frequency signals, analysts have effectively put forward several strategies such as codebook mapping [4-6] consolidating with the Gaussian mixture model (GMMs) [1], and the hidden Markov models (HMMs) for classification with codebooks confronted the disadvantage of low learning capacity [2]. Due to the loss of spectral details, the source-filter-based method keeps degrading speech quality. Recently, more attention has been given to supervised machine learning to map narrow-band speech to wideband speech for improving speech quality. According to research by Hinton et al. [7], using deep neural networks (DNNs) as the acoustic model for HMMs improves phoneme detection over using GMMs, which inspires DNNs to be employed in the speech processing fields like bandwidth extension as classifiers.

Li and colleagues [8] have demonstrated the application of a DNN to learn the relationship between the log-power magnitude of high-frequency and low-frequency components in order to reconstruct the high spectrum by utilizing a mirror version of the phase. However, mirror phase limits the quality of the expanded speech and yet challenging task to enhance speech due to its unstructured characteristics [9]. Meanwhile, a multi-modal speech denoising network obtains noisy speech by concatenating clean speech with a noise source and feeding it to the multi-modal encoder along with the accelerometer signal [10]. The network shows that it is possible to reconstructed extremely good quality outcomes from band-limited signal, however, difficult to deal with different types of background noise in the band-limited signal. In [11], Radio2Speech has been proposed to analyze RF signals to extract high quality speech. The effectiveness of Radio2Speech is equivalent to the microphone in calm situations, according to experimental findings, and continues to function well in noisy and soundproof environments but shows a bit unsatisfied performance with multi-speaker scenario. Previous research by Y. Gu and colleagues utilized a deep recurrent neural network with long short-term memory (LSTM) cells [12] as a bottleneck feature, combined with a

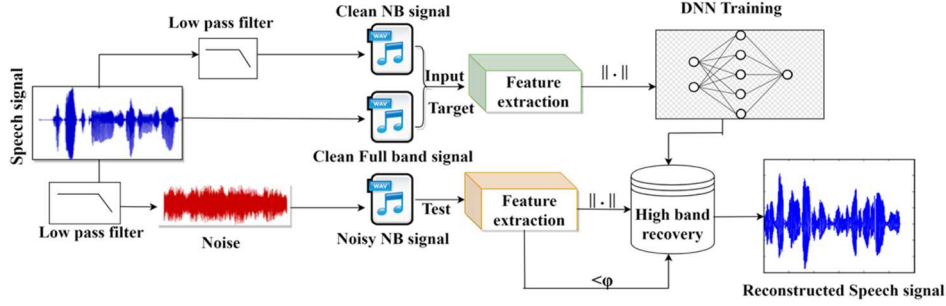


Figure 1. Block diagram of the proposed bandwidth expansion system.

vocoder-based approach using bidirectional input-output mapping [13], for estimating high-frequency spectral components. In this study, we introduce a DNN architecture that can reconstruct clear full-band speech from noisy, band-limited speech signals by learning the relationship between the log-power magnitude of high-frequency and low-frequency components. The DNN has been framed as a regression problem, allowing it to non-linearly model the clean low-frequency components and predict an approximation of the high-frequency components. Given sufficient training data, DNNs can learn complex models, broadening their use in speech recognition and enhancement [2]. The specific goals of this study are twofold: (i) to reduce noise and produce clean speech from noisy speech, and (ii) to expand the bandwidth and recover the high-frequency spectrum.

The paper is organized as follows. Sec. II explains the methodology of the proposed bandwidth expansion networks. It also includes the feature extraction technique leveraged in the proposed network. Experimental setup and results of the proposed DNN based network is described in Sec. III. We compare results of the proposed network with results of the traditional methods. Finally, Sec. IV concludes this study.

II. EXPERIMENTAL SETUP

A. Preprocessing

Basic block diagram of the proposed system is shown in Fig.1. The model is divided into two stages: training and testing stage. The input full band audio signal is passed through 20th-order Butterworth filter with a cutoff frequency of 3000 Hz, resulting a narrowband signal. The STFT, $Y(m, n)$ of the signal is then extracted from the output as follows.

$$Y(m, n) = \sum_{k=0}^{N-1} y(m * \Delta + k)h(k)e^{-2\pi jkn/N} \quad (1)$$

Here, m is the frame index, $m = 0, \dots, M-1$ is the frequency index, Δ is the window shift, N is the window length and $h(k)$ is the Hamming window function Log-spectral power magnitude and phase information has been extracted from STFT as features.

In training stage, the input to the model is STFT magnitude of clean narrowband signal and the target is STFT magnitude

of clean full band audio signal. For testing, noisy narrowband is taken as input to recover clean full band signal.

B. Model Description

The deep neural network (DNN) illustrated in Fig.1 consists of an input layer, an output layer, and three dense layers serving as hidden layers, each with 256 nodes. The DNN receives the log spectrum of the narrowband signal as input and outputs the log spectrum of the full-band signal. To standardize the input and output pairs, they were normalized with a zero mean and unit variance across all training samples. The narrowband signal was processed using a window size of 256 with a window shift of 128. The initial learning rate was set to 0.01, and the network was trained using the Adam optimization method. The mean squared logarithmic error (MSLE) loss function was used to measure the difference between the input and target and ensure effective learning by the proposed system. The MSLE loss function can be defined as,

$$E_i = \sum_{i=1}^n (\log(y_i + 1) - \log(\hat{y}_i + 1))^2 \quad (2)$$

Here n is the number of input signals, y is a data point of the original signal, and $\hat{y}$ is the data point value of the predicted signal.

C. Signal reconstruction

Even though the precise magnitude of the wideband spectra could be obtained, the phase information was lost. Several algorithms have been developed for synthesizing time domain signal, which necessitates the estimate of phase information for high frequencies whose magnitude has been predicted. The high-frequency spectra have been calculated based on DNN output, where certain energy losses are compensated for the energy loss owing to just half of the points of wideband signal is utilized to construct the narrowband spectrum [3]. For reconstructing phase, three conditions have been considered: "Oracle phase" $\angle Y = \angle \hat{Z}$; "Imaged phase" $\angle \hat{Z} = -\text{flip} \angle X$, and "Noisy phase" $\angle X_{\text{noisy}} = \angle \hat{Z}$. After estimating the magnitude of full-band signal, time-domain waveform is reconstructed.

The inverse short-time Fourier transform (ISTFT) has been performed on $\hat{Z}$, overlapping, and adding with the same hamming window to reconstruct the signal.

$$\hat{Z} = \hat{Z}_{\text{mag}} * \exp(j \cdot \angle \hat{Z}) \quad (3)$$

D. Test corpus

TIMIT dataset consists of broadband recordings of 630 speakers of 4000 samples of 8 major dialect regions of the United States. This study used a part of the dataset (1000 samples from different male and female speakers) for training and 50 samples for testing the network. The testing and training speakers are different and 20% of the training speakers are taken for validation purposes. The speech was upsampled to 16 KHz from its original sampling rate. In this study clean signals are used for training the bandwidth extension model. Noise may be a more prominent thought having the most impact on speech signals. Considering such a fact, both clean and noisy signals have been used for testing the model to reconstruct the wideband signals. Two types of noises: speech shape noise and car noise are added along with several SNR values with the clean signal to create a noisy environment. We calculated the scores of objective metrics like perceptual evaluation of speech quality (PESQ) [14], short-time objective intelligibility (STOI) [15], and spectrogram orthogonal polynomial measure (SOPM) [16].

III. RESULTS

The performance of the proposed system is described in this section. The performance scores of different objective metrics are shown for a clean and robust condition with two SNRs [5 dB and 10 dB] respectively.

A. Bandwidth expansion for clean condition

Table I presents the results of an ablation study conducted on reconstructed signal of the TIMIT corpus in a clean scenario. The study compares the original signal with three different reconstructed signals with different phases (Oracle, Imaged, and Noisy). The results show that the reconstructed signal with the oracle phase has the best performance in terms of both PESQ and SOPM, with a score of 4.44 and 0.98 respectively. The imaged phase and the noisy phase have similar performance with scores of 4.44 and 0.97 for PESQ and 0.98 and 0.96 for SOPM. The STOI results show a slightly lower performance for the noisy phase with a score of 0.96 compared to the oracle and imaged phases, which have a score of 0.98.

Table I. Ablation study on the reconstructed signal with three different phases in clean scenario with TIMIT corpus.

Objective Measures	Original Signal	Reconstructed Signal		
		Oracle phase	Imaged phase	Noisy phase
PESQ	4.42	4.44	4.44	4.44
STOI	0.98	0.98	0.98	0.96
SOPM	0.95	0.98	0.97	0.96

B. Bandwidth expansion for noisy condition

Two types of noises are used in this study for assessing the robustness of the proposed system. One is speech-shaped noise (SSN) as full-band noise, and the other is low-band car noise. Table II and Table III show the performance scores of different objective metrics of reconstructed signals in noisy scenario.

Table II. Quantitative evaluation on the reconstructed (Enhanced) signal in a full-band noisy environment.

Objective Measures	10 dB				5 dB			
	Noisy signal	Oracle	Enhanced signal	Noisy	Noisy signal	Oracle	Enhanced signal	Noisy
PESQ	2.30	2.60	2.32	2.32	1.97	2.31	1.97	1.97
STOI	0.86	0.90	0.86	0.87	0.77	0.84	0.79	0.78
SOPM	0.92	0.96	0.93	0.92	0.85	0.90	0.86	0.86

Table II and Table III appear to be the results of an evaluation of audio signal processing. The data is organized into two tables, each presenting result for a different type of noisy environment (full-band or low-band). Each Table presents three objective measures, and these are the standard metrics to evaluate the quality of speech signals. For each measure, the table represents results for four different conditions: the noisy

Table III. Quantitative evaluation on the reconstructed (Enhanced) signal in a low-band noisy environment.

Objective Measures	10 dB				5 dB			
	Noisy signal	Enhanced signal			Noisy signal	Enhanced signal		
	Oracle	Imaged	Noisy	Noisy	Oracle	Imaged	Noisy	
PESQ	2.32	2.60	2.32	2.32	1.97	2.30	1.97	1.97
STOI	0.86	0.90	0.88	0.88	0.76	0.82	0.78	0.77
SOPM	0.91	0.95	0.93	0.923	0.85	0.90	0.86	0.86

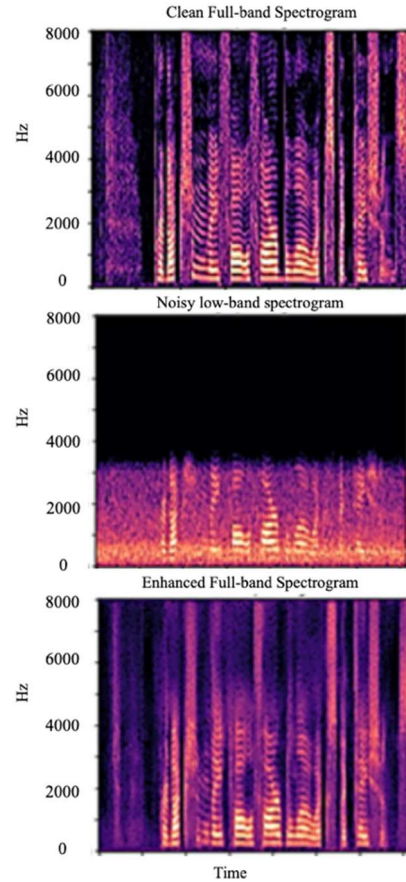


Figure 3. Spectrogram of one test utterance with SNR 10 dB, (a) clean full band signal spectrogram, (b) noisy low band signal spectrogram, (c) enhanced full band signal spectrogram with oracle phase.

signal with low band, an enhanced signal reconstructed by an “oracle” phase, “imaged” phase and “noisy” phase. In general, the results show that the enhanced signal has

higher values for each of the three measures, compared to the noisy low band signal. The difference between the enhanced signal and the noisy signal is more pronounced for the full-band noisy environment, where the enhanced signal has improved PESQ, STOI, and SOPM values by 0.3, 0.04, and 0.04 respectively, compared to the noisy signal at 10 dB, and by 0.34, 0.07, and 0.05 respectively at 5 dB. For the low-band noisy environment, the enhanced signal has improved PESQ, STOI, and SOPM values by 0.28, 0.04, and 0.02 respectively, compared to the noisy signal at 10 dB, and by 0.33, 0.06, and 0.05 respectively at 5 dB. Fig. 3 represents the log magnitude spectrogram analysis of one test utterance among 50 test samples. The proposed study does better in reconstructing the high band from the noisy low band without any discontinuity between the low and high band spectrogram. The spectrogram of the enhanced figure is shown in the bottom of Fig. 3.

IV. CONCLUSION

In conclusion, the paper presents a deep neural network-based approach for joint bandwidth expansion and speech enhancement. The proposed approach aims to increase the bandwidth of speech signals while enhancing their quality and reducing noise. The network employs three different phases along with magnitude spectra to recover high-band components and is designed to simultaneously estimate the missing speech components and the noise profile in the degraded speech signal. The results show that the DNN-based bandwidth extension and speech enhancement approach outperforms in noisy environment. The results of the ablation study and the evaluation of the audio signal processing demonstrate the effectiveness of the enhanced signal compared to the noisy low-band signal. The reconstructed signal with the "oracle" phase performed the best in terms of PESQ and SOPM, with a score of 4.44 and 0.98 respectively. However, the "imaged" and "noisy" phases also showed similar performance. The results were more pronounced in the full-band noisy environment, where the enhanced signal improved PESQ, STOI, and SOPM values by 0.04, 0.03, and 0.01 respectively at 10 dB than at 5 dB. For the low-band noisy environment, the improved values were 0.05, 0.02, and 0.03 respectively at 10 dB than at 5 dB. These results highlight the potential of the proposed method to effectively improve the quality of speech signals in noisy environments. Overall, the results suggest that the quality of the reconstructed signal is maintained to a certain extent even when the phases are not perfectly accurate.

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